Exercise 3 : SQL CASE statements

1.
```
SELECT product_name,
       price,
CASE
    WHEN price > 1000 THEN 'Expensive'
    WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
ELSE 'Budget'
END AS price_category,
FROM product;
```

OUTPUT

Product_name	Price	Price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid_range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

2.
```
SELECT customer_name,
       amount,
CASE
    WHEN amount >= 1000 THEN 'High Value'
    WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'
    WHEN amount < 500 THEN 'Low Value'
END AS order_value_category,
~~Output~~ FROM orders;
```

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2. OUTPUT

customer_name	amount	order_value_category
Alice	150.00	Low_Value
Bob	560.00	Medium_value
Charlie	999.99	Medium_value
Diana	45.50	Low_value
Ethan	1200.00	High_value

3.

```
SELECT emp_name,
       department,
       salary,
   CASE WHEN department = 'IT' AND SALARY > 80000 THEN 'Senior IT'
        WHEN department = 'HR' AND SALARY > 55 000 THEN 'Experienced HR'
        ELSE 'staff'
   END AS position_level,
FROM employees;
```

OUTPUT

emp_name	department	salary	position_level
John	IT	85 000	Senior_IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	staff
Lucy	Finance	95 000	staff
Tom	HR	55 000	staff

4. students

```
SELECT student_name,
       score,
CASE
    WHEN score ≥ 90 THEN 'A'
    WHEN SCORE BETWEEN 80 AND 89 THEN 'B'
    WHEN SCORE BETWEEN 70 AND 79 THEN 'C'
    WHEN SCORE BETWEEN 60 AND 69 THEN 'D'
    ELSE 'F'
END AS grade,
FROM students
```

OUTPUT:

Student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5. deliveries

```
SELECT delivery_id,
       delivery_time_minutes
CASE
    WHEN delivery_time_minutes ≤ 30 mins THEN 'Fast'
    WHEN delivery_time_minutes BETWEEN 31 AND 60 mins THEN
                                                 'ON Time'
    ELSE
    ~~WHEN~~ 'Late'
END AS performance,
FROM deliveries
```

↓

output 5

delivery_id	delivery_time_minutes	perfomance
1	45	'On Time'
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6. tickets

```
SELECT issue_type,
       Priority,
CASE
    WHEN Priority = 3 THEN 'High'
    WHEN Priority = 2 THEN 'Medium'
    ELSE 'Low'
END AS Priority_label,
FROM tickets;
```

output

issue_type	Priority	Priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7. attendance

```
SELECT student_id,
        ((days_present / total_days) * 100) AS attendence_percentage,
CASE
    WHEN ((days_present/total_days)*100) >=90 THEN 'Excellent'
    WHEN ((days_present/total_days)*100) BETWEEN 75 AND 89 THEN 'Good'
    ~~WHEN ((~~
    ELSE 'Needs Improvement'
END AS attendance_status,
FROM attendance
```

output

student_id	~~days_present~~ attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Need improvement
5	100	Excellent

8. Products_inventory

```
SELECT Product_id,
        stock_qty,
CASE
    WHEN stock_qty =0 THEN 'Out of Stock'
    WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
    ELSE 'In Stock'
END AS stock_status,
FROM products_inventory
```

output 8

product_id	stock qty	stock status
1	5	~~Low stock~~ In stock.
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low stock

9. Classes

```
SELECT subject,
       enrolled_students,
CASE
    WHEN enrolled_students >= 25 THEN 'Large'
    WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
    ELSE 'Small'
END AS class_size_Category,
FROM classes;
```

output:

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10' payments.

```
SELECT payment_id,
       payment_method,
       amount,
CASE
    WHEN payment_method = 'cash' AND amount ≥ 200 'Eligible for
                                                   Discount'
    ELSE 'NOT Eligible'
END AS discount_eligibility,
FROM payments;
```

output:

payment_id	payment_method	amount	discount_eligibility
1	card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	Paypal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount